

Complete MTK Hydropower SMIB Model

Refractoring based on state-of-the art work flow

OpenHPLjl development note

20 September 2026

Abstract

This document records the package commit that turns the OpenHPLjl component catalogue into a complete reduced hydropower single-machine infinite-bus (SMIB) model in ModelingToolkit.jl. The model couples hydraulic storage and water-column dynamics, turbine power conversion, shaft inertia, synchronous-generator angle dynamics, an infinite grid, frequency measurement, and a droop governor. The implementation follows a layered workflow: physical law, connector semantics, reusable component, hierarchical composition, structural compilation, consistent initialization, numerical simulation, and verification.

1 Commit objective

The objective of this commit is to move from individually tested components to one connected nonlinear plant model that can be compiled and simulated as a single ModelingToolkit system. The model is deliberately reduced, but every major energy domain required for governor and frequency-response studies is represented explicitly.

The top-level model is exported as `ReducedSMIB`. It is intended to be the baseline system for subsequent guide-vane disturbance tests, governor tuning, AGC extensions, FCR studies, operating-point linearization, Bode/Nyquist analysis, and plant-specific parameterization.

2 System architecture

The physical and control architecture is

Upstream reservoir

|

v

Headrace

|

v

Surge tank

|

v

Penstock

|

v

Shaft-coupled turbine ---> Shaft ---> SMIB generator ---> Infinite grid

^

|

|

v

+--- Droop governor <- Frequency sensor

The system contains eleven named subsystems:

Subsystem	Role
upstream	constant-head upstream boundary
headrace	hydraulic inertia / momentum balance
surge	local hydraulic storage
penstock	downstream water-column inertia
turbine	hydraulic-to-mechanical conversion
tailwater	constant-head downstream boundary
shaft	rotating inertia and damping
generator	SMIB electromechanical conversion
grid	infinite-frequency electrical boundary
frequency_sensor	non-loading speed measurement
governor	first-order droop control

3 Connector organization

The implementation separates physical connectors from causal control signals.

3.1 Hydraulic connector

The hydraulic interface uses head as the potential variable and volumetric flow as the conserved flow variable:

$$H \quad Q.$$

The sign convention is flow positive into a component. Connected hydraulic flow variables therefore sum to zero.

3.2 Rotational connector

The mechanical interface uses

$$\omega \quad \tau,$$

where angular speed is the potential variable and torque is the conserved flow variable.

3.3 Electrical power connector

The reduced grid interface uses

$$\omega_e \quad P,$$

where electrical angular frequency is the across variable and active power is the conserved flow variable.

3.4 Signal connectors

Control and measurement signals use causal connectors rather than physical flow/potential semantics:

```
SignalPlug  ->  SignalSocket
  output      input
```

This allows the turbine gate command to be driven by a governor, AGC block, test signal, or other controller without changing the turbine physics.

4 Hydraulic model

4.1 Headwater and tailwater

The two reservoir boundaries prescribe hydraulic head,

$$H_{up} = 120 \text{ m}, \quad H_{down} = 20 \text{ m}$$

for the default case, while the connected network determines boundary flows.

4.2 Rigid water columns

Both headrace and penstock use

$$\dot{Q} = \frac{gA}{L} (H_{in} - H_{out} - h_f(Q)).$$

The complete-model baseline uses `:none` friction so that the default operating point can be constructed exactly before additional hydraulic-loss complexity is introduced.

4.3 Surge tank

The surge tank follows

$$A_s \dot{H}_s = Q_{in} + Q_{out}.$$

At the nominal condition, equal steady flow passes through the headrace and penstock, so $\dot{H}_s = 0$.

5 Turbine and shaft coupling

The new `ShaftCoupledTurbine` closes the hydraulic-mechanical link.

Turbine head is

$$H_t = H_{in} - H_{out},$$

guide-vane opening is supplied by a causal input,

$$y = u_{gate},$$

and the baseline flow law is

$$Q = K_q y \sqrt{H_t}.$$

Mechanical power is

$$P_h = \rho g \eta_t Q H_t.$$

With the package convention that torque is positive into a component, the generating turbine exports mechanical power through

$$\tau_t \omega = -P_h.$$

The shaft supplies the rotational inertia:

$$J \dot{\omega} = \tau_{drive} + \tau_{load} - D(\omega - \omega_0).$$

6 Reduced SMIB generator

The new `SMIBGenerator` adds a rotor-angle state relative to the infinite bus:

$$\dot{\delta} = \omega_m - \omega_g.$$

Mechanical and electrical power satisfy

$$P_m = \tau_g \omega_m,$$

$$P_e = \eta_g P_m,$$

and the reduced infinite-bus transfer relation is

$$P_e = P_{max} \sin \delta.$$

Generated electrical power leaves the generator component:

$$P_{grid} = -P_e.$$

Rotational inertia remains in `LumpedShaft`; it is not duplicated in the generator.

7 Measurement and governor loop

The `FrequencySensor` reads shaft speed without mechanically loading the shaft:

$$\tau_{sensor} = 0,$$

and reports

$$f = \frac{\omega}{2\pi}.$$

The droop governor computes

$$y_{cmd} = y_0 + \frac{f_{ref} - f}{R},$$

and applies first-order actuator/governor dynamics:

$$T_g \dot{y} = y_{cmd} - y.$$

The complete feedback path is therefore

$$\omega \rightarrow f \rightarrow \text{droop governor} \rightarrow y \rightarrow Q \rightarrow P_h \rightarrow \omega.$$

8 Consistent default operating point

The default values used by `ReducedSMIB` are:

Quantity	Symbol	Default
Upstream head	H_{up}	120 m
Tailwater head	H_{down}	20 m
Initial flow	Q_0	10 m ³ /s
Gate opening	y_0	1.0
Grid frequency	f_0	50 Hz
Turbine efficiency	η_t	0.90
Generator efficiency	η_g	0.98
Generator transfer limit	P_{max}	10 MW
Surge area	A_s	500 m ²
Governor time constant	T_g	0.2 s
Droop coefficient	R	2.5 Hz/pu gate

The initial turbine head is

$$H_{t0} = H_{up} - H_{down} = 100 \text{ m.}$$

The turbine coefficient is constructed from the requested nominal flow,

$$K_{q0} = \frac{Q_0}{y_0 \sqrt{H_{t0}}}.$$

Initial turbine power is

$$P_{h0} = \rho g \eta_t Q_0 H_{t0} = 8.829 \text{ MW.}$$

Initial electrical power is

$$P_{e0} = \eta_g P_{h0} = 8.65242 \text{ MW.}$$

The initial rotor angle is computed from

$$\delta_0 = \sin^{-1} \left(\frac{P_{e0}}{P_{max}} \right),$$

instead of being selected as an arbitrary guess.

At the default condition the governor sees 50 Hz and returns $y = y_0$, water-column flows are equal, the surge level is stationary, and mechanical/electrical power are balanced.

9 Top-level ModelingToolkit composition

The complete system is assembled from components and `connect` statements. The top-level system does not rewrite the internal equations of individual components.

```
@named plant = ReducedSMIB()

compiled = mtkcompile(plant)

prob = ODEProblem(
    compiled,
    [],
    (0.0, 5.0),
)

sol = solve(
    prob,
    Rodas5P();
    abstol = 1e-8,
    reltol = 1e-8,
)
```

The workflow deliberately preserves two model views:

- the hierarchical uncompiled system for component, port, signal, and future analysis-point inspection;
- the compiled system for structural simplification, initialization, and numerical solution.

10 Files introduced by the commit

The principal new implementation files are:

```
src/Measurements/FrequencySensor.jl
src/Controls/DroopGovernor.jl
src/Turbines/ShaftCoupledTurbine.jl
src/Electrical/Generators/SMIBGenerator.jl
src/Systems/ReducedSMIB.jl
examples/reduced_smib.jl
```

The package entry point is updated to include and export these components, and `test/runtests.jl` is extended with a complete plant compile-and-simulate test.

11 Verification strategy

The commit follows a verification ladder rather than only checking whether Julia code parses.

Component verification

Existing tests continue to check reservoir mass balance, rigid-pipe momentum response, surge-tank storage, friction laws, shaft acceleration, and generator algebra.

Composition verification

The top-level model checks that all eleven named systems are present and connected through public interfaces.

Structural compilation

The hierarchical plant is passed through `mtkcompile`, transforming the connected equations into a reduced numerical system.

Numerical equilibrium test

The complete model is solved and checked near

$$f = 50 \text{ Hz}, \quad Q = 10 \text{ m}^3/\text{s}, \quad H_s = 120 \text{ m}, \quad y = 1.0,$$

with

$$P_e \approx 8.65242 \text{ MW}.$$

12 What this commit enables

The package now has one equation-based nonlinear model spanning hydraulics, mechanics, electrical dynamics, measurement, and primary control.

The immediate study sequence is:

1. apply a gate, reference, or load disturbance after a steady interval;
2. inspect headrace and penstock flow transients;
3. inspect surge-tank head oscillation;
4. inspect turbine mechanical and generator electrical power;
5. inspect shaft frequency and governor response;
6. add actuator limits and transient droop;

7. introduce plant-specific parameters and turbine characteristic maps;
8. add analysis points for linearization and frequency-domain analysis;
9. extend the control hierarchy toward AGC and FCR studies.

13 Current model boundaries

This is a reduced baseline rather than the final high-fidelity plant model. Planned extensions include hydraulic losses at the operating point, turbine hill-chart interpolation, gate saturation and rate limits, elastic penstock/water hammer, detailed synchronous-machine voltage/reactive-power dynamics, and plant-specific governor structure.

The value of this commit is that these refinements can now be added one subsystem at a time without changing the complete-system workflow.

14 Conclusion

This commit establishes the first complete ModelingToolkit hydropower SMIB in OpenHPLjl. It converts the package from a collection of reusable physical components into a connected, initialized, compilable, and simulatable nonlinear plant model with closed-loop droop control. The architecture is now suitable as the base for transient studies, control development, system identification, FCR workflows, and later plant-specific validation.